EE322: Embedded Systems Design - Project

Home Security System

Project Proposal

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Home Security System

Introduction

Security is an essential component of modern life, especially in urban and suburban settings where the possibilities of unauthorised entrance, fire threats, and property destruction are high. With advances in embedded systems and sensor technologies, developing an efficient and intelligent home security system has become more possible and cost-effective.

This project proposes a smart home security system to improve domestic safety by combining modern sensors and assembly-level programming to provide efficient and precise control. The system is intended to handle three primary security concerns: unauthorised human mobility, fire or smoke dangers, and glass-breaking sounds. It can detect motion and distinguish between residents, intruders, and pets by combining ultrasonic and PIR sensors based on item height and heat signatures. Smoke detectors and temperature sensors give early warning of a fire or smoke, allowing for quick reaction to threats. In addition, microphone-based sound sensors detect forcible entry by identifying high-frequency patterns such as broken glass. This complete method seeks to provide solid security for modern dwellings.

The suggested system is intended to provide a complete, customizable, and cost-effective solution suitable for residences of various sizes and security requirements. Its revolutionary design strives to bridge gaps in standard security systems by employing intelligent object detection and event-driven alarms, providing homeowners with immediate awareness and enhanced peace of mind.



Figure 01: Model of home security of system

Problem Identification

The demand for an enhanced home security system arises from various shortcomings in existing systems, which this project seeks to fill. Traditional security systems frequently fail to differentiate between authorised and unauthorised users, resulting in numerous false alarms. This is especially difficult in households with pets or several family members. Furthermore, many systems lack dependable procedures for detecting fire dangers and forced entry, such as glass-breaking incidents. Furthermore, commercial smart devices are often internet-dependent, which raises privacy concerns while boosting expenses. This concept aims to address these challenges by incorporating RFID tags to identify authorised residents, as well as ultrasonic and PIR sensors to differentiate motion. The system also incorporates smoke detectors and microphone-based sensors for comprehensive threat detection, ensuring a cost-effective, efficient, and robust solution tailored to modern security needs.

Proposed Solution

To address the stated issues, the suggested solution combines modern sensor technology with assembly programming for a smart home security system. The technology will employ RFID tags to distinguish between authorised home members and intruders. The motion detection system will use a combination of PIR sensors to detect heat signatures and ultrasonic sensors to measure item height, ensuring reliable recognition of unauthorised humans while disregarding pets. Smoke detectors and temperature sensors will be used to provide early notice of a fire or smoke, reducing false alarms from non-hazardous sources. A microphone-based sound sensor will also detect high-frequency sounds patterns such as glass-breaking to identify attempted forceful entry. All sensors will communicate with an assembly-language-programmed microcontroller, allowing for optimal performance and real-time reaction. This system prioritizes cost-efficiency, privacy, and offline functionality, making it robust and suitable for modern security need.

Market Analysis

In Sri Lanka, there is an increasing demand for low-cost, high-efficiency home security systems, particularly in cities and suburbs. Many present options are either prohibitively expensive or depend on internet connectivity, making them inaccessible to middle-income homes. Furthermore, frequent power and connectivity outages in some areas provide difficulties for systems that require continuous online operation. This research deals with these restrictions by providing a low-cost, offline-capable solution that combines RFID for user identification with motion, smoke, and sound sensors for comprehensive security. This system meets the needs of the Sri Lankan market while also addressing worldwide trends in home security technology by prioritizing privacy, cost, and usability.

Implementation (Methodology)

The methodology for this project involves a systematic approach to designing and implementing a smart home security system. Initially, the requirements are analysed to define the system's objectives, including motion detection, fire or smoke identification, and glass-breaking sound recognition. Suitable hardware is then selected, such as ultrasonic and PIR sensors for motion detection and, for Authorized vs. Unauthorized members (home members and others) we planned to implement RFID tag for authorized members. Smoke detectors for hazard alerts, and microphone-based sensors for sound recognition.

These sensors are integrated with a microcontroller, like the 8051 or AVR, programmed in assembly language to ensure efficient, low-level control. Algorithms are developed to process sensor data, such as differentiating between humans and pets, detecting smoke or abnormal temperatures, and identifying specific sound patterns. The system is prototyped and calibrated for optimal sensitivity, followed by rigorous testing to ensure reliable operation under various conditions. After deployment in a controlled environment, performance is evaluated, and refinements are made based on testing outcomes and user feedback. Finally, the system is documented and presented, showcasing its functionality and advantages over existing solutions.

Sensors for motion detection



Figure 02: PIR Sensor



Figure 03: Ultrasonic sensor



Figure 04:RFID Module & tag

Fire smoke detection



Figure 05: DSM501A Sensor

Sound Detection Sensor



Figure 06: KY-037 Sensor

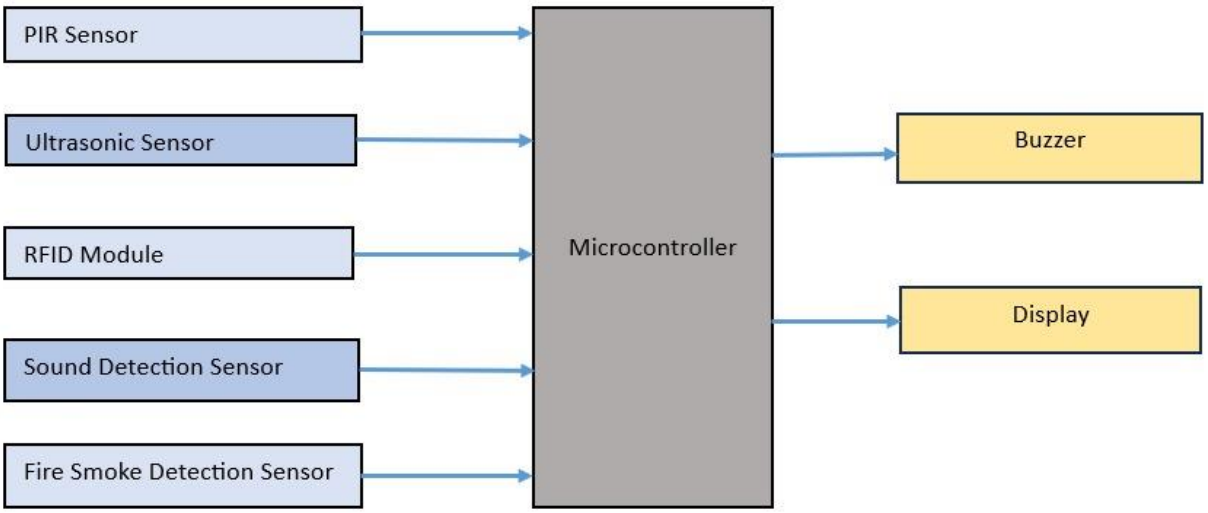


Figure 07: Suggested Design

Cost Analysis

Description	*Unit price/ Rs*	*Quantity*	*Total / Rs*
Arduino uno	2600.00	1	2600.00
PIR sensor	300.00	1	300.00
Ultrasonic sensor	325.00	1	325.00
RFID Module	380.00	1	380.00
RFID tags	60.00	5	300.00
DSM501A sensor	1700.00	1	1700.00
KY -037 Sensor	240.00	1	240.00
Resistors	20.00	5	100.00
Buzzer Module	140.00	1	140.00
Breadboard	380.00	1	380.00
Jumper Wires and wires	-	-	300.00
16x2 LCD Display	470.00	1	470.00
Other expenses	-	-	500.00
Total			7735.00

TABLE 01: Approximated cost analysis

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